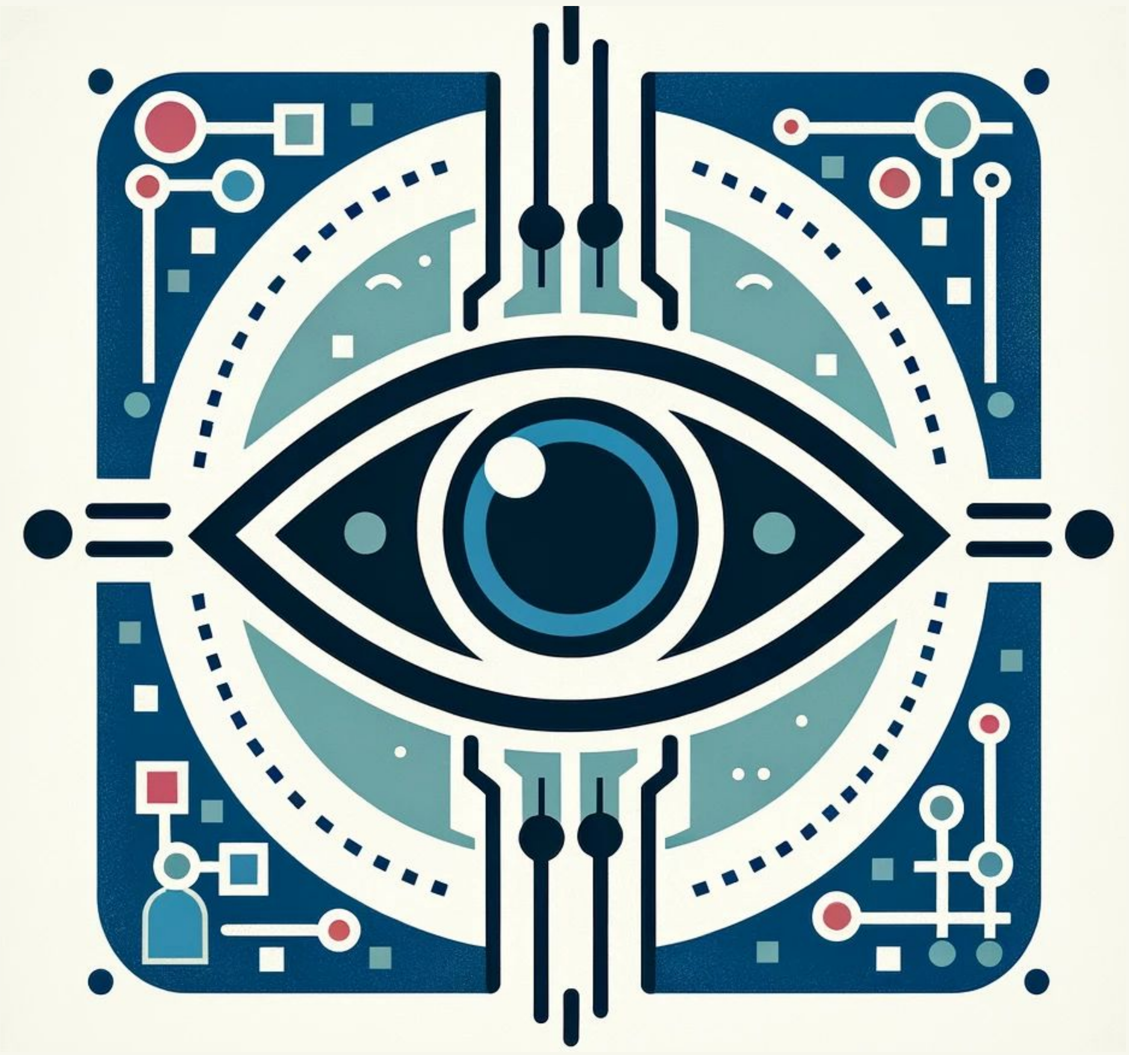


COMPUTER VISION

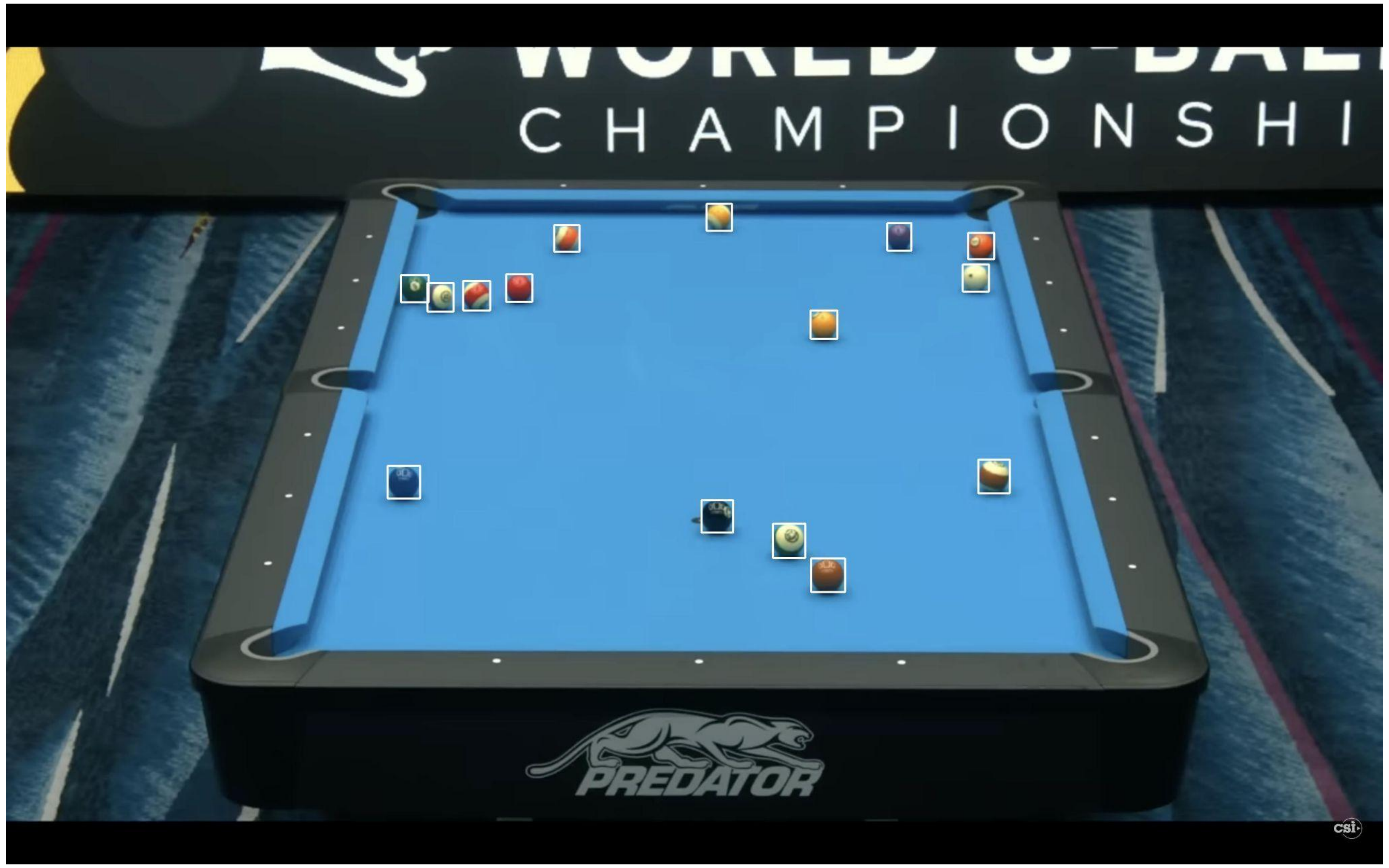
Project


1
14
10
3

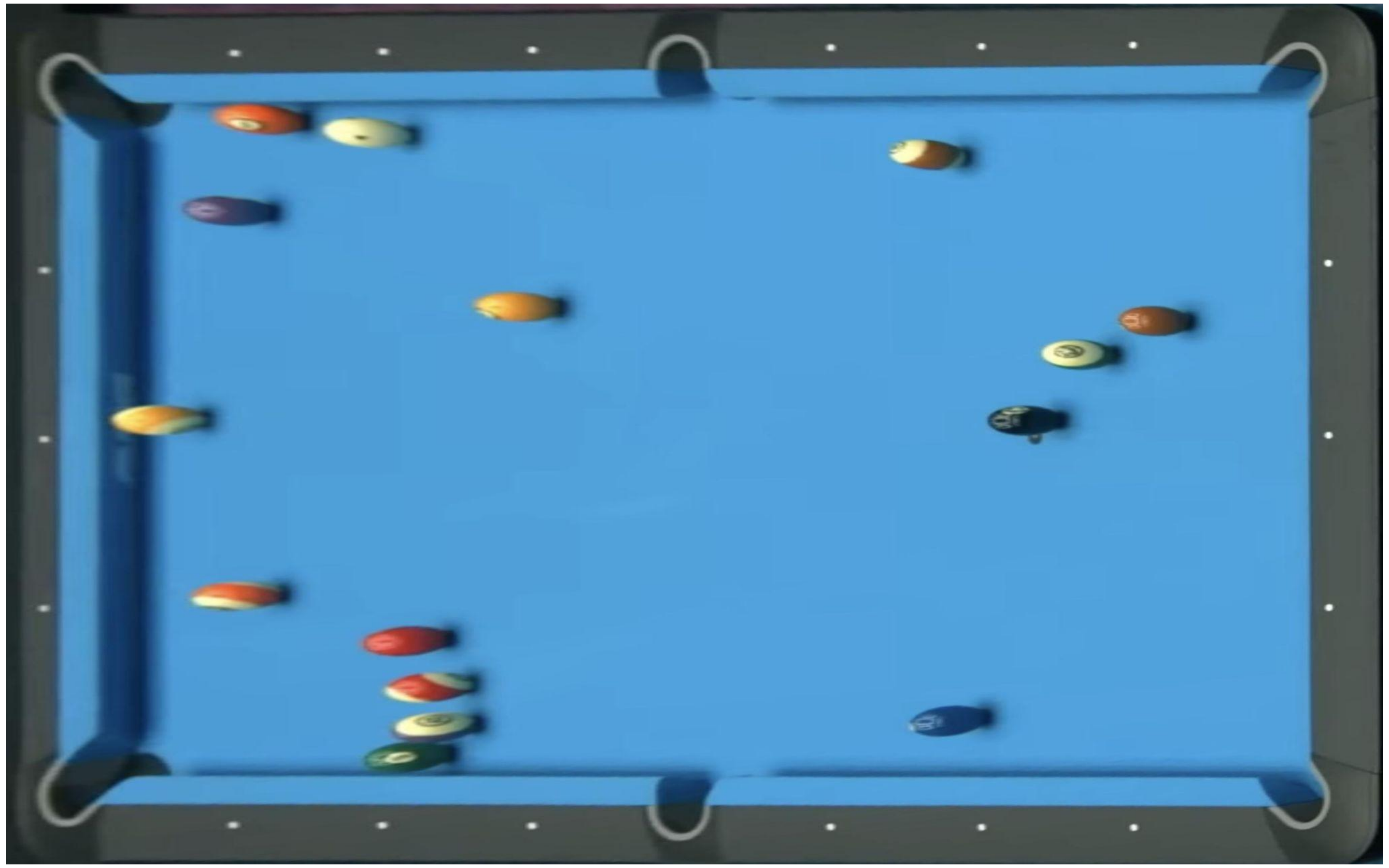
TASK 1

- Input:
  - Image of 8-ball pool table
- Output:
  - Total number of balls on the table
  - Position of the balls on the image (bounding boxes)
  - Number of each ball
  - Top-view of the table

TASK 1



Bounding Box Detection



Top-View of the Table

Dataset: https://universe.roboflow.com/bachelorthesis/8-ball-pool-l530o

TASK 1

- Ball numbers can be obtained based on their colour
- Consider the cue (white) ball to be zero

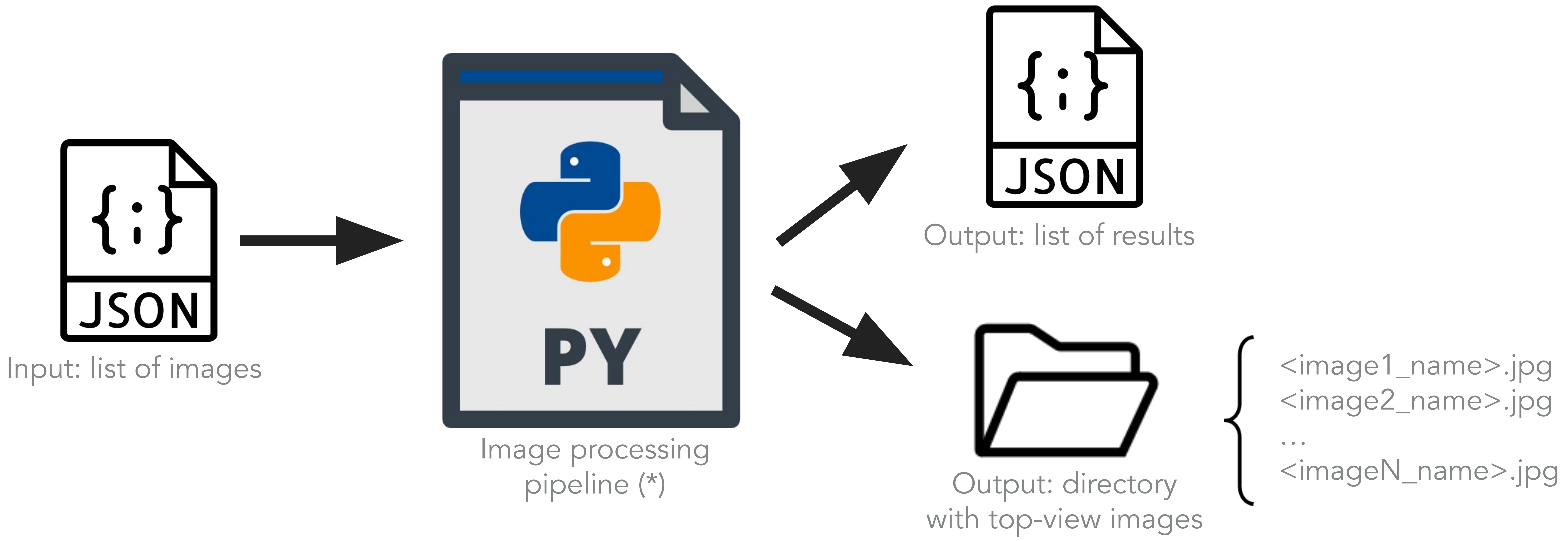
TASK 1

- Dataset:
  - 50 images randomly chosen from a public dataset
  - The results will be tested in 10 **undisclosed** images

- Deliverables:
  - Short report (2 pages max) presenting the **methodology** and some **results**
  - Python script (only one file)

- Deadline: **April, 17** (23:59 AoE)

TASK 1



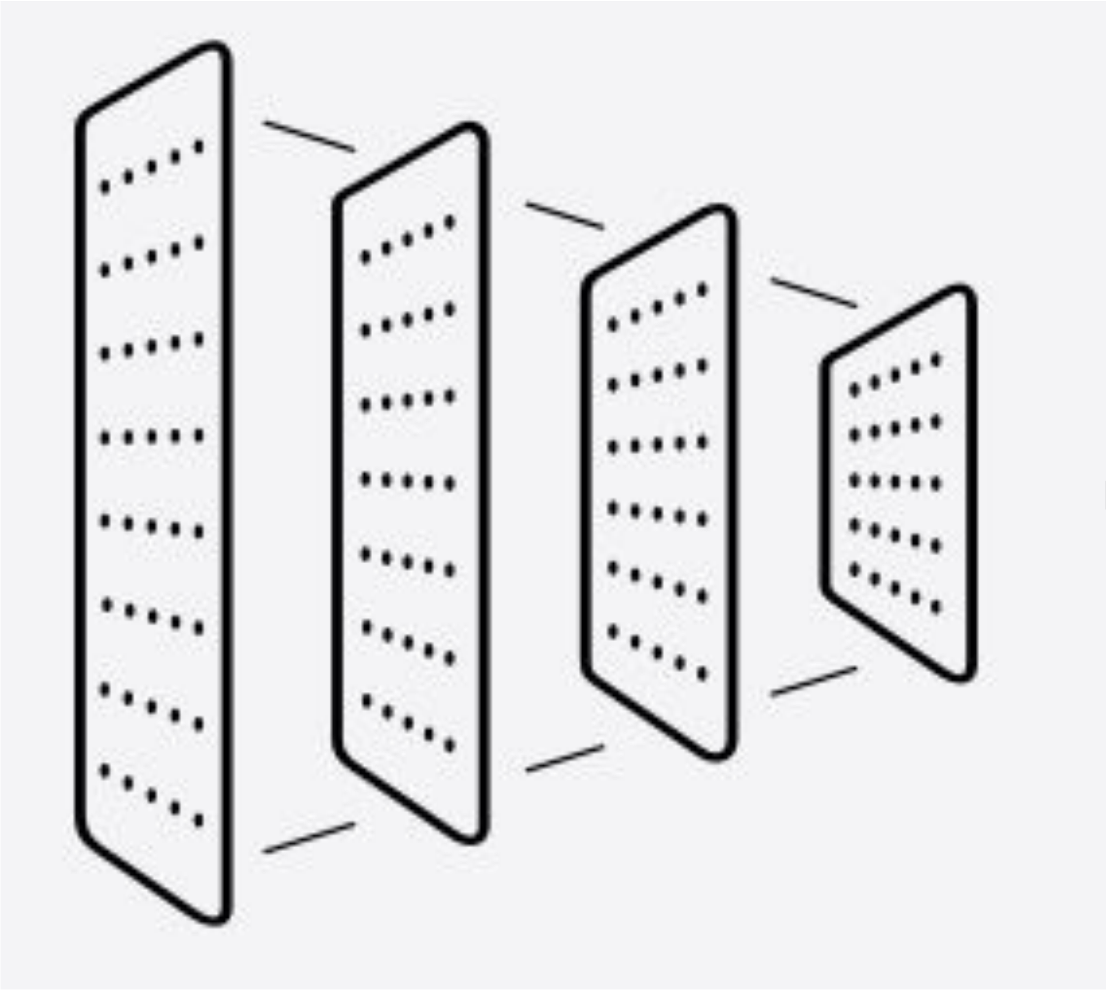
(*) using only OpenCV and other common libraries, like numpy and matplotlib

TASK 1

- Grading
  - Task 1 accounts for **40%** of the overall project grade
  - Elements being considered: methodology, report and quality of the results

- Important remarks
  - Follow **strictly** the JSON structure for the input and output files
  - It is **okay** to use AI tools while developing your work, but it is **not okay** to use them without acknowledging it
  - All members of the group **are expected to understand** the methodology and the submitted code

TASK 2



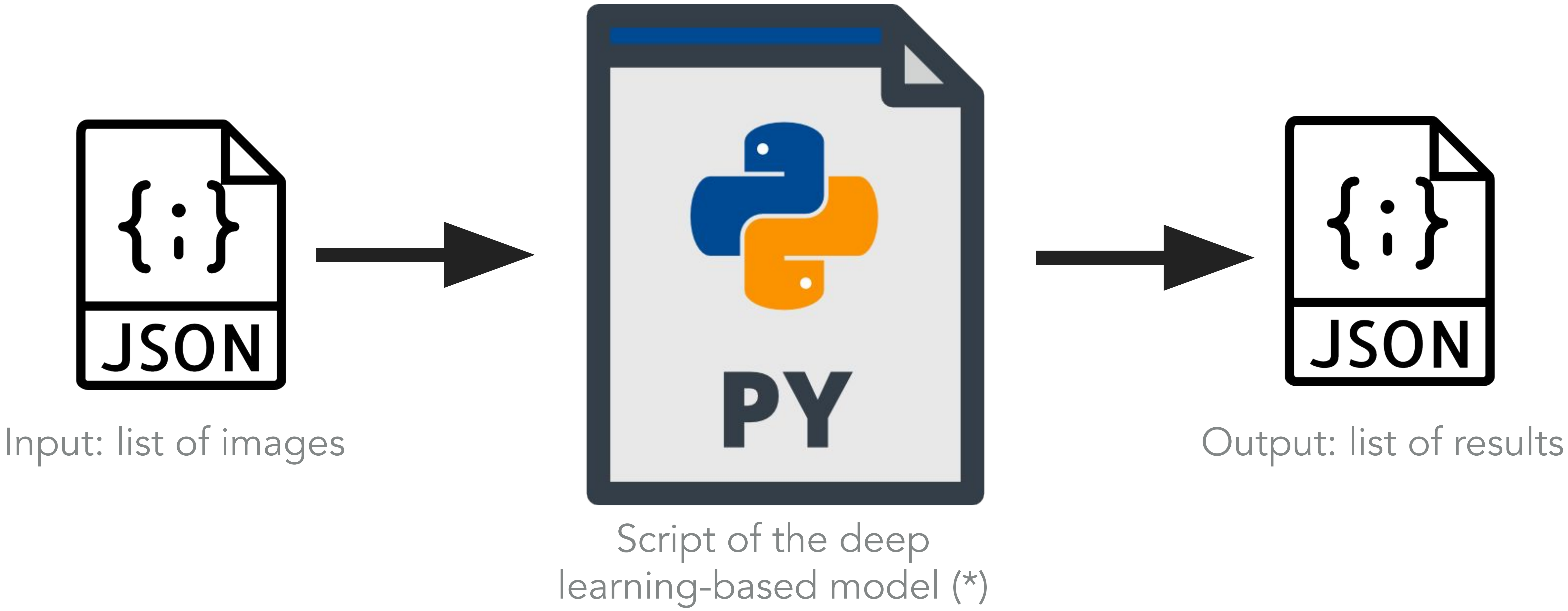
15

CNN image by Oleksandr Panasovskyi

TASK 2

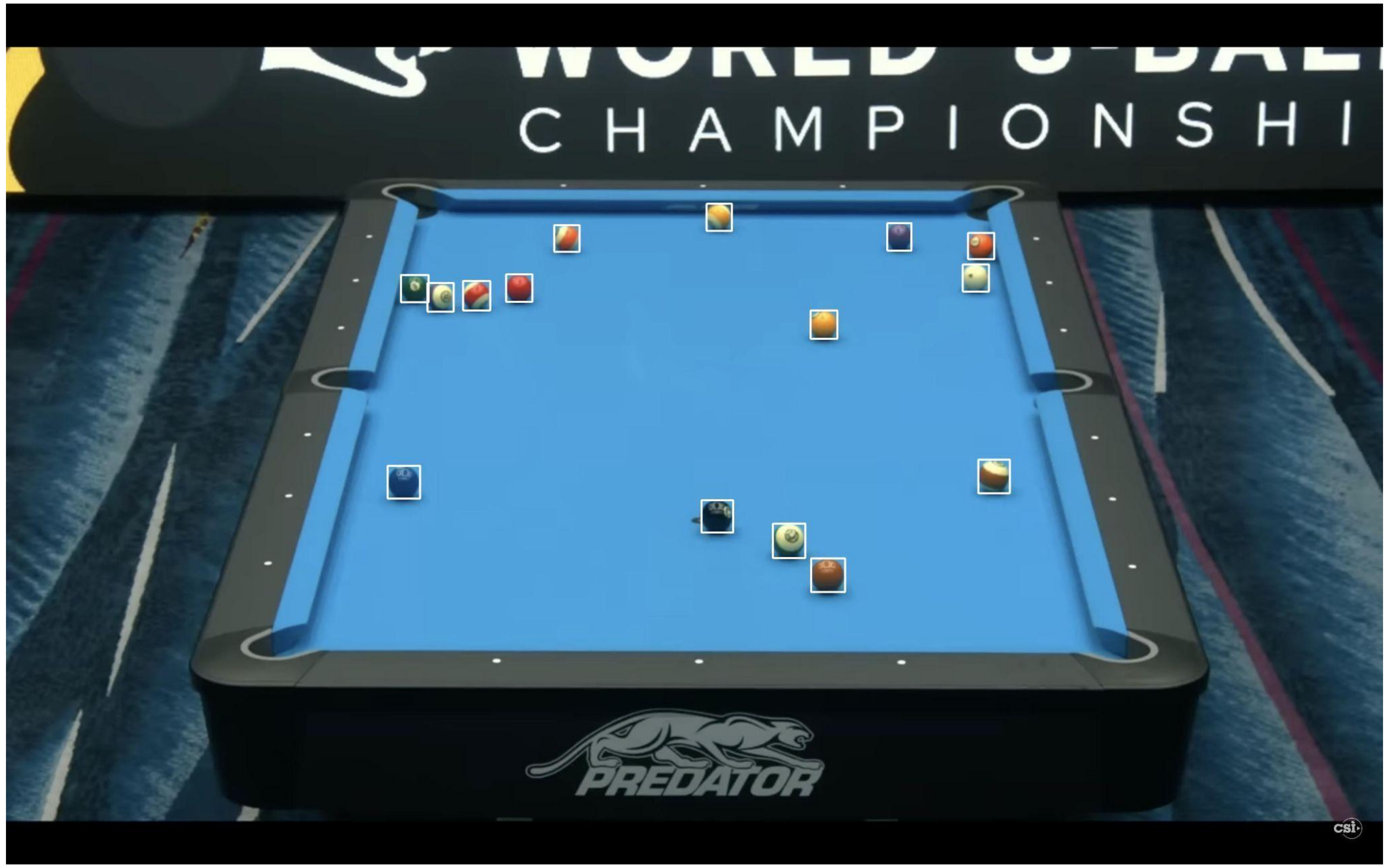
- Input:
  - Image of 8-ball pool table
- Output:
  - Total number of balls on top of the table
- Model(s):
  - CNN-based architecture
  - extra: quantitative comparison (with **adequate metrics**) different architectures

TASK 2



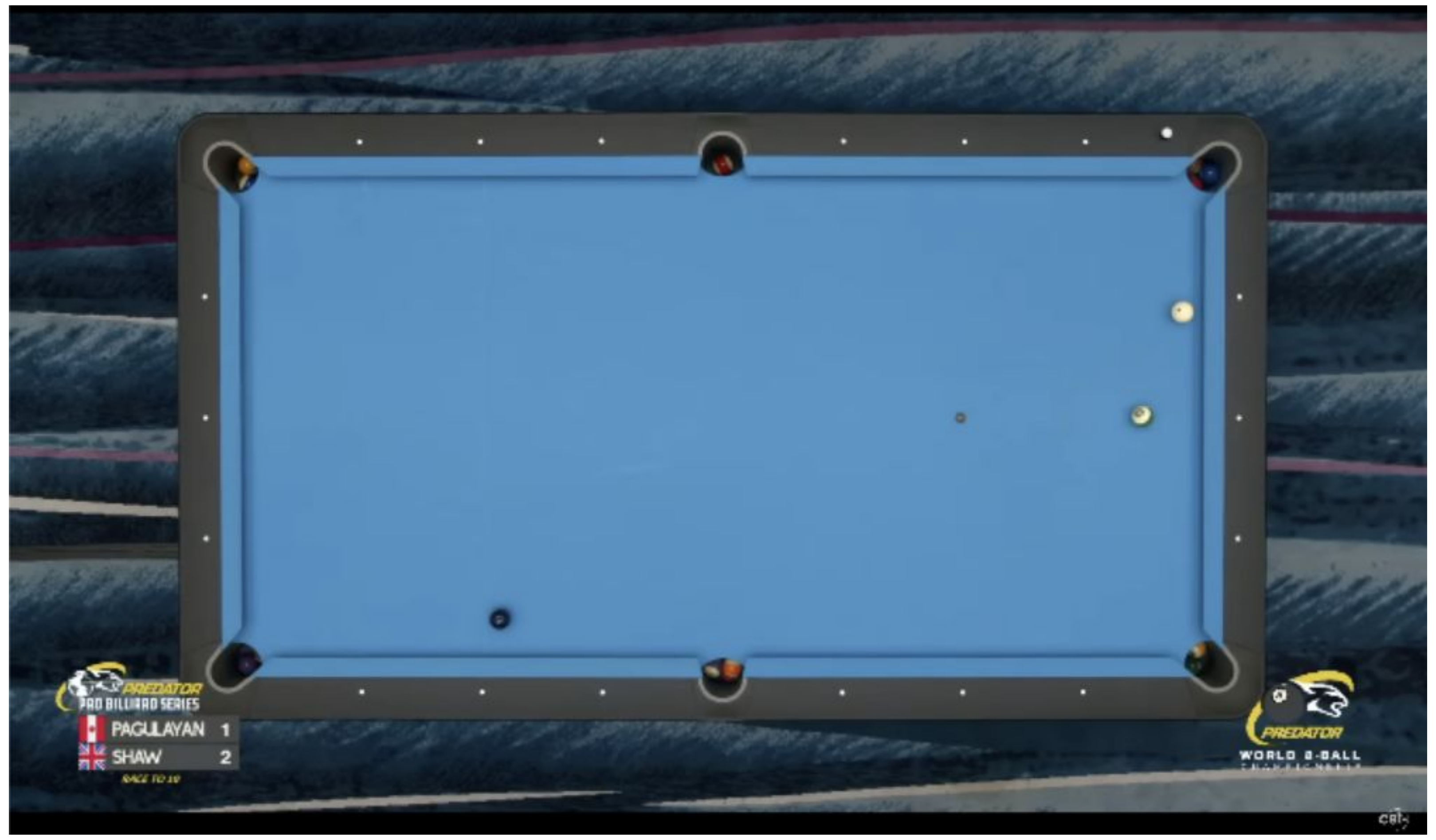
(*) using only PyTorch, OpenCV and other common libraries, like numpy and matplotlib

TASK 3



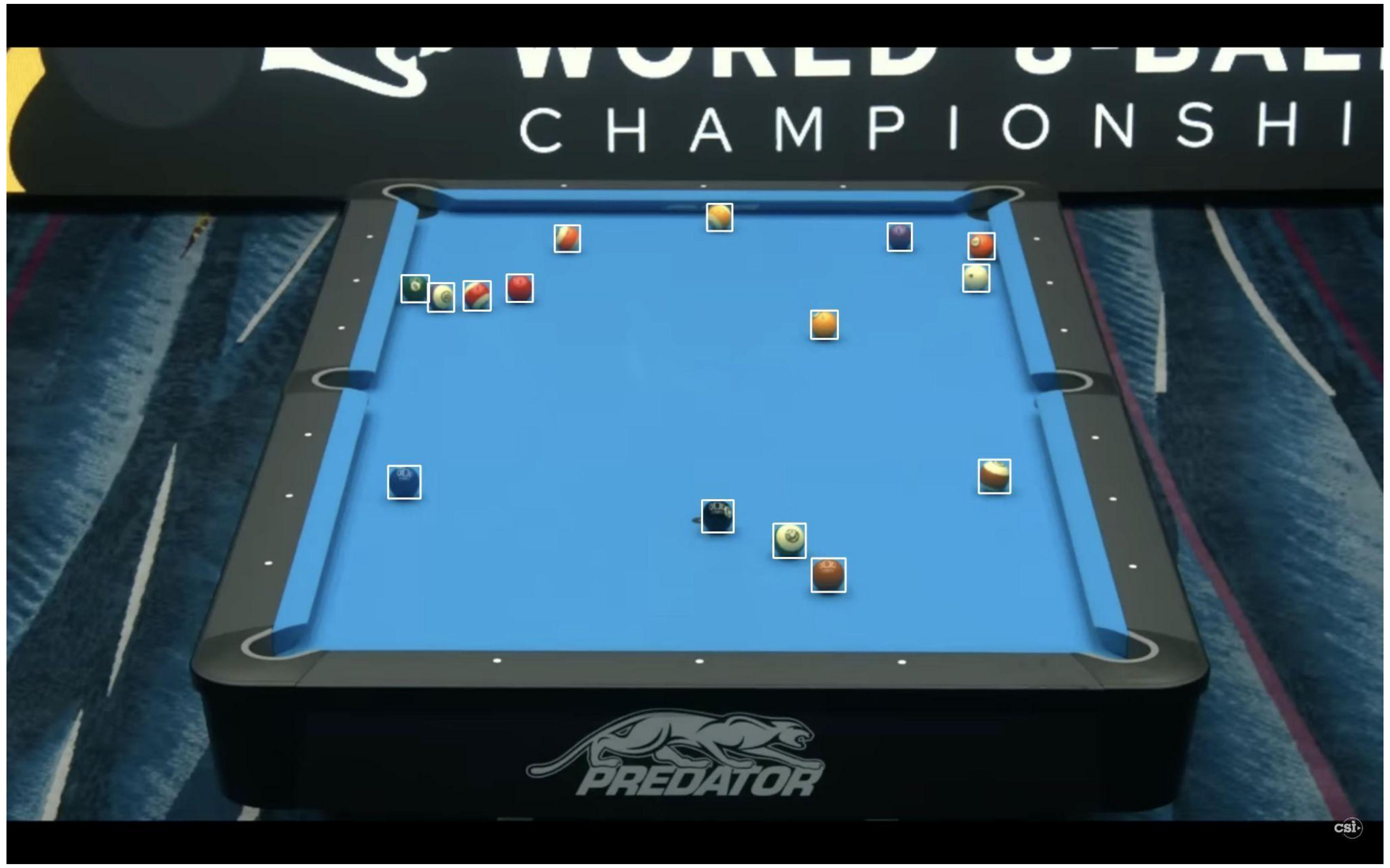
Ball Detection

+



Retrieval

TASK 3



- Ball Detection:
  - At least one model, e.g. YOLO, DETR
  - extra: quantitative comparison (with **adequate metrics**) of different architectures
- Table Retrieval:
  - Select the most similar pooling table from the training data
  - Qualitative evaluation with some (good and bad) results are enough

TASKS 2 + 3

- Main Dataset:
  - https://universe.roboflow.com/bachelorthesis/8-ball-pool-l530o
  - Use train/validation/test splits of the dataset

- Examples of other datasets that can additionally be used for training (should be documented on the final report):
  - https://universe.roboflow.com/nidacorian-protonmail-com/pool-billiard
  - https://universe.roboflow.com/mark-dj0yk/pool-balls-detection-srlqi
  - https://universe.roboflow.com/pool-ball-detection/pool-ball-detection-61fd9
  - others

TASKS 2 + 3

- Deliverables:
  - Short report (3 pages max) presenting the **methodology** and some **results**
  - Python script (only one file) + model (.pth) for <u>task 2</u>
  - Notebook with some results for <u>task 3</u>
- Deadline: **June 22** (23:59 AoE)
- Presentation: **TBD**

TASKS 2 + 3

- Grading
  - Tasks 2+3 account for **50%** of the overall project grade
  - Elements being considered: methodology, report and quality of the results
  - The remaining 10% results from the presentation

- Important remarks
  - Follow **strictly** the JSON structure for the input and output files
  - It is **okay** to use third-party code and AI tools while developing your work, but it is **not okay** to use them without acknowledging it
  - All members of the group **are expected to understand** the methodology and the submitted code